
Phantom Phases: How Attention Sinks Generate Fictitious Training Dynamics

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Abstract

Standard spectral analysis of transformer training has revealed a rich narrative of geometric phases, rank collapse, and representational reorganization. We offer an alternative explanation: these phenomena are generated by a single confound—the attention sink—that progressively dominates the very metrics used to observe it. A one-parameter model of sink contamination predicts the observed E_1 trajectory with $R^2 = 0.978$, and removing the sink direction (*de-sinking*) eliminates all apparent phase transitions. Injecting a synthetic sink into clean representations reproduces the phantom trajectory from scratch, establishing a complete causal loop. The true training dynamics tell a fundamentally different story: effective rank *grows* ($191 \rightarrow 236$), not collapses ($178 \rightarrow 6.4$); raw effective rank is significantly *anti-correlated* with model capability across models ($\rho = -0.59$ to -0.72 , $p < 0.02$); and inter-layer CKA drops by up to 0.57, revealing layer differentiation hidden beneath a shared artifact. Moreover, de-sinking uncovers a previously invisible phenomenon: hidden-state alignment with the unembedding matrix peaks early in training then declines—a *shortcut-to-full-rank transition* observed across Pythia (70M–410M, all peaking at step 512) and OLMo-2 1B (step 1,000). We quantify contamination across five metrics, models from 70M to 6.9B parameters, and both GPT-NeoX and GQA architectures, finding distortions up to $721\times$.

1 Introduction

Li et al. [2025] recently identified three non-monotonic geometric phases in transformer pretraining: an initial representational collapse, a period of dimensionality expansion, and a final compression phase correlated with downstream capability. Their finding, published at NeurIPS 2025, synthesized a growing body of evidence from five independent spectral metrics—PCA variance concentration (E_1), effective rank, RankMe, CKA, and anisotropy—each independently suggesting that transformer training proceeds through abrupt geometric reorganizations [Garrido et al., 2023, Ethayarajh, 2019, Raghu et al., 2021, Timkey & van Schijndel, 2021]. The convergence of multiple metrics on the same narrative has been taken as strong evidence that these phases are real properties of training dynamics.

We show that this convergence has a simpler explanation. All five metrics share a single blind spot: the *attention sink* [Xiao et al., 2024], an input-invariant, high-norm component at position 0 whose magnitude grows monotonically during training. Because this one component dominates the principal spectrum, any metric sensitive to variance concentration will track *sink growth* rather than representational change—producing coherent but fictitious structure. The apparent convergence across metrics is not independent confirmation; it is five measurements of the same artifact.

This claim is testable. If geometric phases are driven by sink contamination rather than representation dynamics, then (1) a contamination model with sink strength as the sole variable should predict the observed E_1 trajectory; (2) removing the sink direction should eliminate all phase transitions;

and (3) the decontaminated metrics should tell a qualitatively different story. All three predictions hold. A single-parameter model fits E_1 with $R^2 = 0.978$ (Theorem 1). De-sinking eliminates Phases 2–3 entirely. And the decontaminated effective rank shows monotonic *growth* ($191 \rightarrow 236$), the opposite of the collapse ($178 \rightarrow 6.4$) reported by raw metrics. Moreover, injecting a synthetic sink into clean representations reproduces the entire phantom trajectory from scratch (§3.4), completing the causal loop: the attention sink is both necessary and sufficient to produce the observed spectral artifacts. Raw effective rank is not merely noisy but *actively misleading*: it is significantly anti-correlated with probing accuracy across models ($\rho = -0.59$ to -0.72 , $p < 0.02$), suggesting the model is deteriorating at exactly the steps where it is learning most rapidly.

De-sinking does more than correct existing measurements—it reveals new ones. With the dominant artifact removed, we observe a previously invisible training phenomenon: hidden-state alignment with the unembedding matrix peaks early in training then declines. This *shortcut-to-full-rank transition* occurs at step 512 across all three Pythia model sizes (70M/160M/410M) and at step 1,000 in OLMo-2 1B, suggesting an intrinsic property of transformer optimization: models first learn a low-rank shortcut (the marginal token distribution), then depart toward richer, full-rank representations.

Contributions.

1. **Sink contamination** (§3–4): five standard spectral metrics are systematically biased by the attention sink. Rogue dimensions $\approx$ sink direction (95.7% overlap). The effect is specific to PC1. A single-parameter contamination model achieves $R^2 = 0.978$. Injecting a synthetic sink into clean representations reproduces the phantom trajectory, completing the causal loop.
2. **Phantom phases dissolved** (§5): de-sinking eliminates geometric phases and reverses rank collapse to rank growth across three model sizes (70M/160M/1.4B). Raw ER is significantly anti-correlated with capability ($\rho = -0.59$ to -0.72 , $p < 0.02$).
3. **A hidden training transition revealed** (§6): unembedding alignment peaks then declines—universal across four model families, with a depth-dependent inverted-U departure profile.
4. **Scale and architecture validated** (§4): contamination spans GPT-NeoX and GQA architectures (Qwen2.5: sink α up to $113\times$, spectral gap $154\times$) and worsens with model size, reaching $721\times$ ER distortion at 6.9B.

2 Background and Related Work

Attention sinks. Xiao et al. [2024] identified disproportionate attention to the first token; Darcet et al. [2024] proposed register tokens for ViTs. Sinks arise from the softmax constraint [Sun et al., 2024], emerge early and strengthen with scale [Gu et al., 2025], and serve computational roles with near-zero value states [Barbero et al., 2025]. Cancedda [2024] decomposed sink signals via the unembedding spectrum. Zhang et al. [2025] showed 1–2 sink “tags” explain 20–70% of PCA variance in attention outputs. Queipo-de-Llano et al. [2025] proved massive activations *necessarily* collapse effective rank, interpreting this as functional compression. Their Theorem 1 is *static* (sinks collapse ER at a given layer); our Theorem 1 establishes the *dynamic* consequence: phantom phase transitions across training checkpoints. They study what sinks *are*; we study what sinks *do to training-dynamics measurements*.

Spectral training dynamics. Li et al. [2025] identified three geometric phases using RankMe/ α -ReQ on OLMo and Pythia. E_1 /anisotropy: Ethayarajh [2019]; rogue dimensions: Timkey & van Schijndel [2021]; RankMe: Garrido et al. [2023]; CKA: Kornblith et al. [2019]. Kulkarni et al. [2026] trained 108 models and found effective rank primarily reflects hyperparameters (batch size, weight decay)—a *different* confound from ours. None of these works controls for the attention sink.

Unembedding alignment. The logit lens [nostalgebraist, 2020] and tuned lens [Belrose et al., 2023] established that alignment varies across *layers* in trained models. No prior work tracks alignment across *training time*.

Simplicity bias. Chang & Bergen [2022] showed models acquire words in frequency order; Rende et al. [2024] demonstrated transformers learn interactions from low to high order. These support interpreting our alignment peak as a shortcut phase.

Algorithm 1 De-sinking

Require: Hidden states $\mathbf{H} \in \mathbb{R}^{n \times d}$
1: $\bar{\mathbf{H}} \leftarrow \mathbf{H} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \mathbf{H}$ {Center}
2: $\mathbf{U}, \Sigma, \mathbf{V}^\top \leftarrow \text{SVD}(\bar{\mathbf{H}})$ {Thin SVD}
3: $\mathbf{s} \leftarrow \mathbf{V}_{:,1}$ {Sink direction}
4: $\mathbf{H}_{\text{ds}} \leftarrow \mathbf{H} - (\mathbf{H}\mathbf{s})\mathbf{s}^\top$ {Project out}
5: **return** $\mathbf{H}_{\text{ds}}$

3 The Attention Sink Contaminates Spectral Metrics

3.1 De-Sinking

Definition 1 (De-sinking). Let $\mathbf{H} \in \mathbb{R}^{n \times d}$ be hidden states. Let $\mathbf{s} \in \mathbb{R}^d$ be the first right singular vector of the centered $\mathbf{H}$. The de-sinked representations are $\mathbf{H}_{\text{ds}} = \mathbf{H} - (\mathbf{H}\mathbf{s})\mathbf{s}^\top$.

Both this method and an alternative (normalized mean of position-0 across sequences) yield nearly identical directions (§8), confirming that PC1 in sink-affected layers is the sink direction.

De-sinking is specific, not generic. De-sinking operationally resembles All-but-the-Top [Mu & Viswanath, 2018] and its extensions [Li et al., 2020, Rajaei & Pilehvar, 2021], but serves a different purpose: it is a *diagnostic correction* for training dynamics analysis, grounded in Theorem 1, not an engineering recipe for downstream performance. The effect is specific to PC1: removing PC2, PC3, or a random direction produces negligible change (Appendix C). The spectral gap between the first and second singular values provides a principled justification for removing exactly one direction: in GPT-2, the gap exceeds $14\times$ at L2–L4, while in Pythia-70M it reaches $4.6\times$ at L2. Non-sink layers show gaps of $1\text{--}2\times$.

3.2 Why Contamination Is Inevitable

The attention sink creates a component that is (i) high-variance, (ii) input-invariant, and (iii) shared across layers. We show contamination follows from (i) alone.

Theorem 1 (E_1 Contamination). Let $\mathbf{H} = \alpha \mathbf{a} \mathbf{s}^\top + \mathbf{C}$, where $\mathbf{s}$ is a unit vector, $\|\mathbf{a}\|^2/n = \sigma_s^2$, and $\mathbf{C}\mathbf{s} \approx \mathbf{0}$. Let $\Sigma_c = \frac{1}{n} \mathbf{C}^\top \mathbf{C}$ have trace V_c . Then:

$$E_1^{\text{raw}} = \frac{\kappa \alpha^2 + E_1^c}{\kappa \alpha^2 + 1}, \quad \kappa = \sigma_s^2 / V_c. \quad (1)$$

As $\alpha \rightarrow \infty$, $E_1^{\text{raw}} \rightarrow 1$ regardless of E_1^c .

Proof sketch. Under orthogonality, $\Sigma = \alpha^2 \sigma_s^2 \mathbf{s} \mathbf{s}^\top + \Sigma_c$. Largest eigenvalue $\lambda_1 \approx \alpha^2 \sigma_s^2$; total variance $\alpha^2 \sigma_s^2 + V_c$. Full proof in Appendix A. $\square$

Corollary 1 (Effective Rank Collapse). As $\alpha \rightarrow \infty$, $\text{ER} \rightarrow 1$ regardless of $\text{ER}(\mathbf{C})$.

Corollary 2 (CKA Homogenization). Layers sharing sink direction $\mathbf{s}$ have $\text{CKA} \rightarrow 1$ as sink strength grows.

Corollary 3 (Anisotropy Inflation). Average pairwise cosine similarity increases with α .

Theorem 1 is not merely diagnostic—it is *predictive*. Given any content matrix $\mathbf{C}$ and sink strength α , it predicts the exact degree of contamination. This means phantom phases can be manufactured from first principles: any training run where sink strength grows monotonically will produce apparent rank collapse in raw metrics, regardless of what the content representations actually do.

3.3 Empirical Validation

We validate on Pythia-70M [Biderman et al., 2023], 20 checkpoints (steps 0–143,000). Sink strength grows from $1.06\times$ at step 32 to $10.3\times$ at step 64,000.

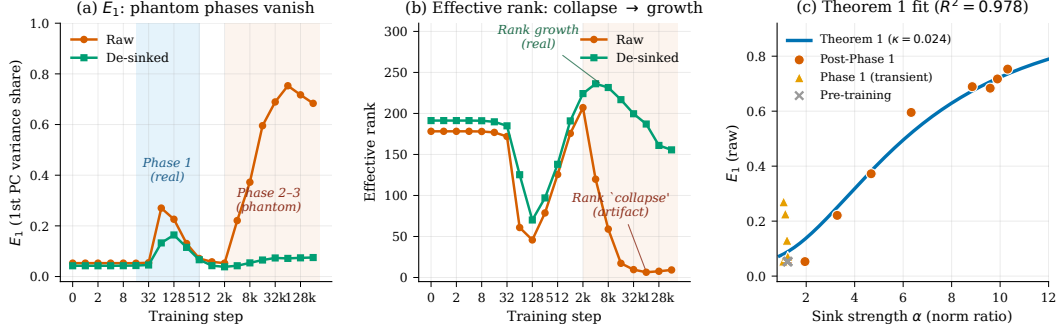


Figure 1. Phantom phases in transformer training (Pythia-70M, L3). (a) Phases 2–3 vanish after de-sinking; only Phase 1 (blue) survives. (b) “Rank collapse” (178 → 6.4) reverses to rank *growth* (191 → 236). (c) Theorem 1 predicts contamination with $R^2 = 0.978$ (one parameter). Phase 1 points (triangles) deviate—content genuinely reorganizes during this period.

Table 1. Contamination audit. Maximum effect across layers.

Metric	Reference	GPT-2 raw	GPT-2 ds	Pythia raw	Pythia ds
E_1 share	Li et al. [2025]	93.6%	11.5%	62.6%	7.0%
Rogue dims	Timkey & van Schijndel [2021]	93.7%	14.6%	—	—
sink overlap		95.7%		—	—
RankMe	Garrido et al. [2023]	352	612	355	415
Anisotropy	Ethayarajh [2019]	0.55	0.48	0.98	0.13
CKA (max $ \Delta $)	Raghu et al. [2021]	−0.569		+0.474	

Figure 1(c) shows the fit: $\kappa = 0.024$ yields $R^2 = 0.978$. Phase 1 transient points ($\sim$ step 32–512) deviate from theory because content representations genuinely reorganize during this period—precisely the behavior expected if Phase 1 is a real training transition and Phases 2–3 are artifacts of sink growth.

Model robustness. Theorem 1 (1 param, $R^2 = 0.978$, AIC = −127.8) outperforms the best generic alternative (logistic, 3 params, $R^2 = 0.945$, AIC = −105.3) because it leverages content information via E_1^c . Leave-3-out cross-validation (20 splits) yields $R^2 = 0.972$; Phase 1 residuals are $1.55\times$ larger than non-Phase-1, consistent with genuine representational change during that period.

3.4 Manufacturing Phantom Phases

The evidence so far establishes *necessity*: removing the sink eliminates phantom structure. We now establish *sufficiency*: injecting a synthetic sink into clean representations produces phantom structure from scratch.

Starting from Pythia-70M at step 512—where the sink has not yet formed ($\alpha \approx 1.0$)—we de-sink the hidden states, then add a rank-1 perturbation along the empirical sink direction with controlled strength α . As α grows from 0 to 30, E_1 rises from 0.063 to 0.898 and ER collapses from 140 to 2.3 (Appendix D)—reproducing the entire phantom trajectory from clean representations. Combined with the removal experiments (§3.3), this completes a causal loop: the attention sink is both necessary *and* sufficient to produce the observed spectral artifacts.

4 Scope of Contamination

4.1 Five Metrics, Two Model Families

We apply de-sinking to GPT-2 [Radford et al., 2019] (124M) and Pythia-70M on 200K tokens from OpenWebText.

The most striking result: the sink direction accounts for 90–96% of rogue dimension variance at L3–L10 in GPT-2. The “rogue dimensions” that Timkey & van Schijndel [2021] identified as ob-

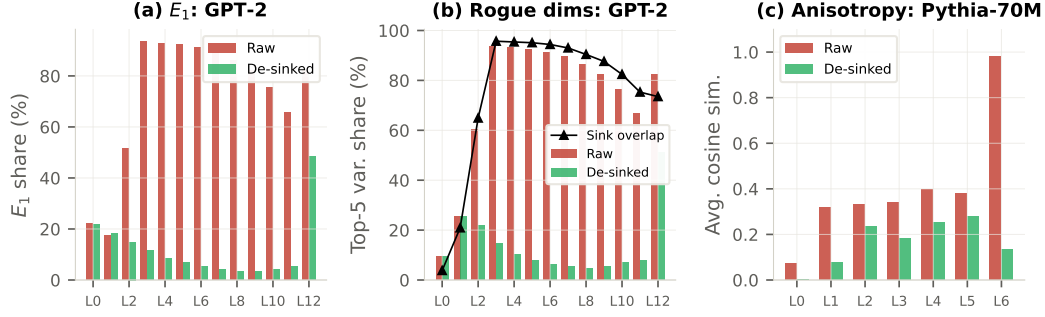


Figure 2. Metric contamination across layers. (a) E_1 in GPT-2: raw 94%, de-sinked 3–12%. (b) Rogue dimensions and sink overlap (>90% at L3–L10). (c) Anisotropy in Pythia-70M: L6 drops from 0.98 to 0.13.

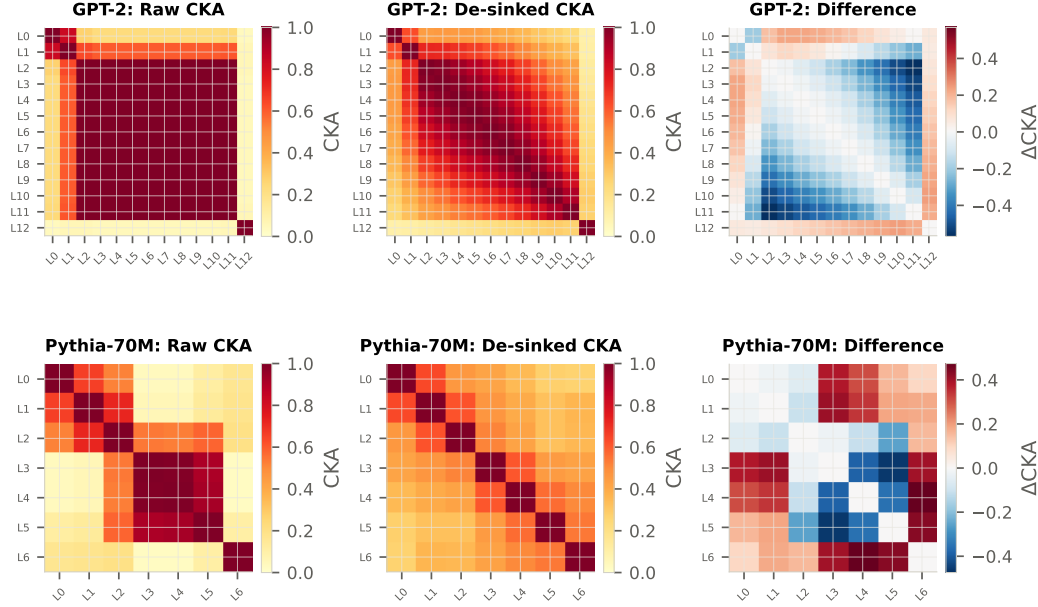


Figure 3. CKA heatmaps. Top: GPT-2. Raw CKA shows L2–L11 as virtually identical (≈ 1.0); de-sinked CKA reveals a rich gradient structure ($\max |\Delta| = 0.569$). Bottom: Pythia-70M.

scoring representational quality *are* the attention sink. This single observation unifies two previously disconnected literatures.

4.2 CKA Reveals Hidden Layer Differentiation

An aggregate CKA comparison suggests minimal impact (GPT-2: $0.703 \rightarrow 0.610$). This masks the truth.

Raw CKA shows L2–L11 as virtually identical (≈ 1.0), consistent with the narrative that deep layers “converge” during training. De-sinked CKA reveals a monotone gradient from L2 to L11 ($\max |\Delta| = 0.569$), consistent with progressive refinement—layers are *differentiated*, not converged (Figure 3).

4.3 Contamination Worsens with Scale

This trend has an uncomfortable implication: the larger the model, the less reliable standard spectral analysis becomes. For Pythia-1.4B, sink strength reaches $14.8\times$ (vs. $10.3\times$ for 70M), and the raw ER collapse is more severe: $23 \rightarrow 2.7$ while de-sinked ER shows growth to 27.2 before plateauing. The pattern extends beyond the GPT-NeoX family: Qwen2.5-1.5B, which uses grouped-query attention (GQA), RoPE, and SwiGLU—the dominant modern architecture—exhibits *even more extreme* contamination, with sink strength reaching $113\times$ and spectral gaps exceeding $150\times$ (Table 2). The

Table 2. Contamination worsens with scale and spans architectures. [†]Full 20-checkpoint training dynamics. [‡]GQA architecture (grouped-query attention, RoPE, SwiGLU).

Model	Params	Max E_1 inflation	Max ER distortion	Peak α	Peak layer
Pythia-70M [†]	70M	16×	30×	10.3×	L3
GPT-2	124M	17×	165×	—	L3
Pythia-160M [†]	160M	6.5×	6.9×	10.4×	L5
GPT-2-XL	1.5B	15×	290×	—	L12
Pythia-1.4B [†]	1.4B	6.6×	8.5×	14.8×	L12
Qwen2.5-1.5B [‡]	1.5B	29×	—	113×	L2
Pythia-6.9B	6.9B	40×	721×	—	L8

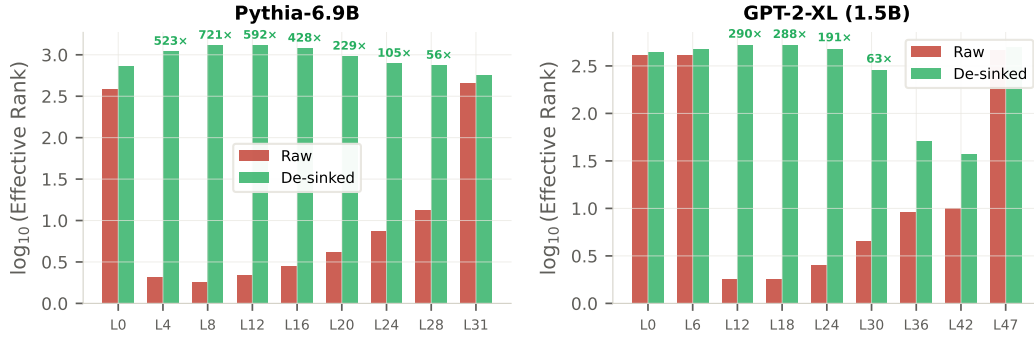


Figure 4. Contamination worsens with scale. ER (log scale) for Pythia-6.9B and GPT-2-XL. Pythia-6.9B L8: 721×

phantom dynamics are not a small-model or old-architecture curiosity—they intensify at scale and generalize across architectures.

5 The True Training Dynamics

5.1 Rank Growth, Not Collapse

Figure 1(b) shows the central positive finding. After de-sinking, Pythia-70M L3 shows: stable ER ≈ 190 (steps 0–32); a genuine Phase 1 dip and recovery (steps 32–512); rank *growth* from 138 to 236 (steps 512–8,000); and gradual plateau. The “rank collapse” from 178 to 6.4 is entirely a sink artifact.

This pattern reproduces across scales. Pythia-160M L5: raw ER collapses from 10.8 to 1.8 while de-sinked ER grows from 9.9 to 12.4. Pythia-1.4B L12: raw ER collapses from 23.0 to 2.7 while de-sinked ER grows from 23.3 to 27.2 before plateauing at 17.9. In all three models, de-sinking reverses the sign of the ER trajectory.

5.2 Raw Metrics Give the Wrong Sign

The contamination does not merely add noise—it reverses the direction of the signal. Over 20 checkpoints, raw effective rank is significantly *anti-correlated* with probing accuracy at sink-dominated layers: Pythia-70M L3 yields Spearman $\rho = -0.59$ ($p = 0.017$); Pythia-160M L3 yields $\rho = -0.72$ ($p = 0.002$). De-sinked ER corrects this directional error (70M: $\rho = +0.25$; 160M: $\rho = -0.33$, n.s.), removing the significant misleading signal in both cases.

A researcher monitoring raw ER would conclude the model is deteriorating at exactly the steps where it is learning most rapidly.

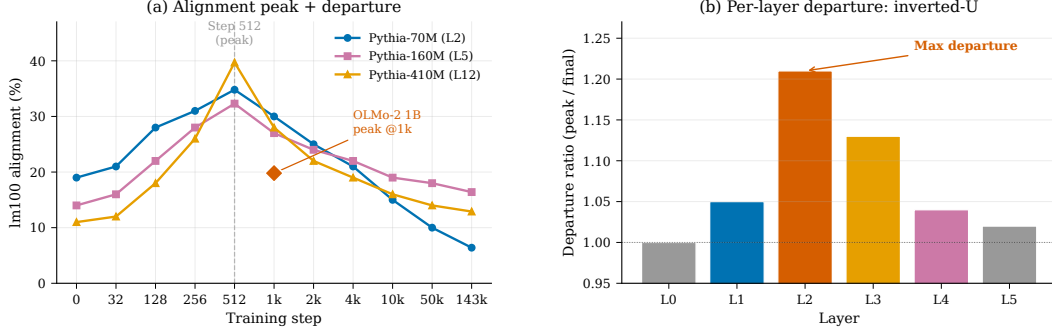


Figure 5. The shortcut-to-full-rank transition. (a) $lm100$ alignment peaks at step 512 across all Pythia sizes, then declines. OLMo-2 1B peaks at step 1,000 (diamond). (b) Per-layer departure ratio follows an inverted-U: middle layers depart most.

182 5.3 Only Phase 1 Survives

183 Phase 1 ($\sim$ step 32–128) appears in *both* raw and de-sinked metrics: de-sinked E_1 shows a transient
 184 spike ($0.04 \rightarrow 0.16 \rightarrow 0.04$), de-sinked ER drops and recovers, and perplexity plummets from
 185 $> 10^6$ to 168. Phases 2–3 exist only in raw metrics and track the growth of sink norm ratio.

186 Of the three geometric phases reported by Li et al. [2025], one is real and two are phantom.

187 6 A Hidden Training Transition

188 De-sinking does more than correct existing measurements—it clears the spectral foreground, mak-
 189 ing subtler geometric signals measurable. We now show that one such signal reveals a previously
 190 invisible training phenomenon.

191 We define *lm100 alignment* as the fraction of hidden-state variance captured by the top-100 singular
 192 vectors of $\mathbf{W}_{lm_head}$.

193 6.1 The Alignment Peak

194 Figure 5(a) shows the trajectory. Early in training, alignment rises sharply as models learn the
 195 marginal token distribution—efficiently solved by aligning hidden states with the unembedding ma-
 196 trix’s high-gain directions. After step 512, alignment declines: the model transitions from this
 197 low-rank shortcut to full-rank encoding.

198 Four observations confirm this is genuine: (i) de-sinked ER expands rapidly at the alignment peak
 199 (Pythia-410M: $30 \rightarrow 49$ at steps 512–1,000); (ii) all Pythia models peak at step 512 despite different
 200 loss values—step-driven, not loss-driven; (iii) OLMo-2 1B (Llama architecture) peaks at step 1,000,
 201 confirming architecture-generalizability; (iv) sink E_1 is only 10.8–13.7% at peak, so the alignment signal
 202 is independent of the sink.

203 6.2 Middle Layers Depart Most

204 The departure profile follows an inverted-U (Appendix E): L0 never aligns strongly (peak 29%) and
 205 cannot depart; L5 is structurally constrained to remain aligned for prediction; middle layers (L2:
 206 $1.21\times$) depart most because they face alignment pressure from the gradient *and* have downstream
 207 layers to compensate.

208 6.3 Interpretation: From Shortcut to Full-Rank Computation

209 Early training aligns hidden states with the unembedding matrix—a low-rank shortcut for the
 210 marginal token distribution [Chang & Bergen, 2022, Rende et al., 2024]. As high-frequency tokens
 211 saturate, the model departs toward full-rank, context-dependent representations. This transition is
 212 invisible in raw metrics because the attention sink dominates the principal spectrum during exactly
 213 this period.

7 Mechanism: Computationally Necessary, Informationally Empty

Causal necessity. Zeroing position-0 hidden states causes perplexity to explode $8\times-107,000\times$.

Informational emptiness. Linear probing shows $|\Delta| < 0.3\%$ at L0–L5. At the final layer, de-sinking *improves* probing by +12.1% (Pythia-70M) and +8.3% (Pythia-160M)—the sink interferes with readout when aligned with the unembedding matrix (cosine 0.97–0.99 in Pythia; 0.26 in GPT-2).

Sink growth $\neq$ capability. Probing accuracy grows monotonically (0.241 $\rightarrow$ 0.352) while sink norm ratio follows a qualitatively different trajectory (Figure 6). De-sinking delta is < 0.015 at every checkpoint.

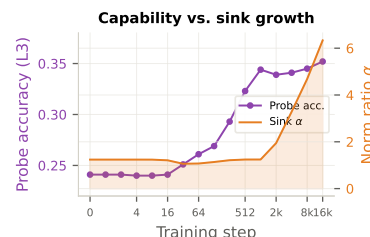


Figure 6. Capability grows monotonically; sink follows a different trajectory. De-sink $\Delta < 0.015$ throughout.

8 Controls

Sink vs. random direction. Removing the sink direction on Pythia-70M L3 reduces E_1 from 0.997 to 0.089; removing a random direction produces negligible change. With 5 random controls per layer, random removal changes probe accuracy by $< 0.05\%$; sink removal yields +12.1% at L6.

Two de-sinking methods agree. PC1 of centered activations and normalized mean of position-0 yield nearly identical directions at sink-affected layers.

PC1 is not the sink at all layers. De-sinking targets layers with large spectral gaps (GPT-2: $> 14\times$ at L2–L4; Pythia-70M: $4.6\times$ at L2). At low-gap layers, PC1 may represent content variance, and de-sinking has negligible effect (Appendix C). We recommend applying de-sinking only where the gap exceeds $\sim 3\times$.

9 Scope, Limitations, and Discussion

Engineering robustness. Contamination distorts *absolute* values but preserves *relative* layer ordering: raw Block Influence pruning (PPL 31.2) outperforms de-sinked (33.8); cross-model CKA is *higher* after de-sinking; distillation is equivalent.

Limitations. Validated on GPT-2, Pythia (70M–6.9B), OLMo, and Qwen2.5 (GQA); de-sinking is linear; alignment peak validated up to 1B; inverted-U profile from a 6-layer model. Sinks serve functional roles [Barbero et al., 2025, Queipo-de-Llano et al., 2025, Zhang et al., 2025], but render training-dynamics *metrics* unreliable [cf. Li et al., 2025, Kulkarni et al., 2026]. After de-sinking, one genuine transition is followed by rank growth—the opposite of collapse. We recommend reporting both raw and de-sinked values: $H_{ds} = H - (H @ s) * s$.

Broader Impact Statement

This work identifies a systematic measurement bias in representation analysis. The positive impact is improved accuracy of scientific claims about how transformers learn. We see no direct negative societal implications.

Reproducibility Statement

All models are publicly available (GPT-2, Pythia, OLMo-2). De-sinking requires $O(nd)$ computation. Anonymized code is available at <https://anonymous.4open.science/r/desink-57F5/> and in the supplementary material; a permanent public release will accompany publication. Total compute: ~ 10 GPU-hours on a single A100.

Use of AI Assistance

Generative AI assistants (Anthropic Claude) were used throughout the research process for ideation, mathematical derivation checking, code implementation, experiment design, data analysis, visualization, and manuscript drafting and editing. All theorem statements, proofs, algorithmic choices, experimental results, figures, and citations were independently validated and finalized by the author, who takes full responsibility for the correctness and originality of the submission.

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301 A Proof of Theorem 1

302 *Proof.* Let $\mathbf{H} = \alpha \mathbf{a} \mathbf{s}^\top + \mathbf{C}$, $\|\mathbf{s}\| = 1$, $\frac{1}{n} \|\mathbf{a}\|^2 = \sigma_s^2$, $\mathbf{C} \mathbf{s} = \boldsymbol{\epsilon}$ with $\|\boldsymbol{\epsilon}\|$ small. The sample covariance
303 is $\boldsymbol{\Sigma} = \alpha^2 \sigma_s^2 \mathbf{s} \mathbf{s}^\top + \boldsymbol{\Sigma}_c + \alpha \mathbf{s} \mathbf{b}^\top + \alpha \mathbf{b} \mathbf{s}^\top$, where $\mathbf{b} = \frac{1}{n} \mathbf{C}^\top \mathbf{a}$. Under the sink model, $\|\mathbf{b}\| \approx 0$
304 (position-0 uncorrelated with content). Setting $\mathbf{b} = \mathbf{0}$ and using $\boldsymbol{\Sigma}_c \mathbf{s} \approx \mathbf{0}$: $\lambda_1 \approx \alpha^2 \sigma_s^2$, total
305 variance $= \alpha^2 \sigma_s^2 + V_c$. Thus $E_1 = \kappa \alpha^2 / (\kappa \alpha^2 + 1)$ where $\kappa = \sigma_s^2 / V_c$. When orthogonality is
306 approximate, $\mathbf{s}^\top \boldsymbol{\Sigma}_c \mathbf{s} = E_1^{c,\parallel} V_c > 0$, yielding (1). $E_1^{\text{raw}} \rightarrow 1$ as $\alpha \rightarrow \infty$. $\square$ $\square$

307 **Empirical verification of near-orthogonality.** The assumption $\mathbf{C} \mathbf{s} \approx \mathbf{0}$ holds exactly by con-
308 struction, since $\mathbf{s}$ is defined as PC1 of the centered matrix. The substantive question is whether the
309 residual content matrix $\mathbf{C}$ has low variance concentration. We verify this across three model scales:
310 $E_1(\mathbf{C})$ remains below 0.07 at all checkpoints (Table 3), confirming the residual spectrum is diffuse
311 and Theorem 1’s decomposition is well-conditioned.

Table 3. Orthogonality verification. $E_1(\mathbf{C})$: leading eigenvalue share of content matrix after de-sinking. p0-conc: fraction of sink projection concentrated at position 0.

Model	Step	$E_1(\mathbf{H})$	$E_1(\mathbf{C})$	p0-conc	Sink α
Pythia-70M L3	512	0.077	0.060	0.3%	$1.2\times$
	4,000	0.281	0.037	14.4%	$3.3\times$
	64,000	0.671	0.056	31.1%	$10.0\times$
	143,000	0.629	0.071	32.5%	$9.6\times$
Pythia-160M L5	512	0.061	0.053	0.2%	$1.1\times$
	143,000	0.829	0.052	43.5%	$18.5\times$
Pythia-1.4B L12	512	0.083	0.068	0.04%	$0.8\times$
	143,000	0.894	0.033	49.8%	$20.3\times$

312 B Extended Probing Results

Table 4. Probing accuracy with/without de-sinking. WikiText-2, 500 seqs, 3 seeds, 5 random controls.

Model	Layer	Sink %	Raw	De-sink	Δ	Random
GPT-2	L3	97.1	39.22	39.21	-0.01	39.22
	L6	95.9	42.57	42.57	+0.00	42.57
	L12	77.7	49.81	50.00	+0.20	49.81
Pythia-70M	L3	64.7	40.22	40.27	+0.05	40.24
	L6	0.4	31.87	42.95	+11.09	31.37
Pythia-160M	L6	90.2	42.50	42.50	+0.00	—
	L12	1.0	38.78	47.08	+8.30	—

313 C PC1 Specificity

Table 5. De-sinking is specific to PC1. Removing PC2, PC3, or a random direction has negligible effect. Pythia-70M, step 143,000.

		Original	Remove PC1	Remove PC2	Remove random
L3	E_1	0.279	0.048	0.304	0.273
	ER	14.6	32.1	14.8	15.7
L5	E_1	0.529	0.142	0.567	0.529
	ER	9.2	25.6	8.3	9.2

D Manufacturing Phantom Phases

We supplement §3.4 with the full sweep of injected sink strengths. Starting from de-sinked Pythia-70M hidden states at step 512 (where the empirical sink has not yet formed), we add a rank-1 perturbation $\alpha \mathbf{e}_0 \mathbf{s}^\top$ with controlled α and re-measure E_1 and ER (Table 6).

Table 6. Injecting a synthetic sink into de-sinked step-512 hidden states (Pythia-70M L3) reproduces the entire phantom trajectory.

Injected α	0	3	5	7	10	15	20	30
E_1	0.063	0.081	0.197	0.325	0.496	0.689	0.797	0.898
ER	140	124	87	53	24	9	5	2.3

E Alignment Peak and Per-Layer Departure

We supplement §6 with the universal alignment-peak data (Table 7) and the per-layer departure profile (Table 8).

Table 7. The alignment peak is universal. All Pythia models peak at step 512; OLMo-2 1B peaks at step 1,000.

Model	Mid layer	Peak step	Peak lm100	Final lm100	Retained
Pythia-70M	L2	512	34.8%	6.4%	18%
Pythia-160M	L5	512	32.3%	16.4%	51%
Pythia-410M	L12	512	39.7%	12.9%	33%
OLMo-2 1B	—	1,000	19.8%	5.5%	28%

Table 8. Per-layer departure: inverted-U. Middle layers depart most; first and last do not.

Layer	Dist. to output	Peak lm100	Final lm100	Departure
L0	5	29.4%	29.3%	1.00×
L1	4	45.8%	43.6%	1.05×
L2	3	62.2%	51.7%	1.21×
L3	2	68.2%	60.3%	1.13×
L4	1	72.4%	69.9%	1.04×
L5	0	77.2%	75.8%	1.02×

F Negative Results

Layer pruning. Pythia-6.9B, 8/32 layers by Block Influence: raw PPL 31.2, de-sinked 33.8. Raw better—sink-heavy layers are genuinely redundant.

LoRA layer selection. Pythia-1B: raw [0,1,3,15], de-sinked [0,2,3,15]. De-sinked diverged. Inconclusive.

Cross-model CKA. GPT-2 vs. Pythia-70M: de-sinked CKA *higher*—similarity genuine.

Distillation. GPT-2 $\rightarrow$ 6-layer: minimal difference between raw/de-sinked/even spacing.

G Full Sink–Unembedding Geometry

Table 9. Sink geometry. Final-layer alignment explains probing differences across model families.

	E_1^{raw}	E_1^{ds}	sink $\leftrightarrow$ lm	Probe Δ
<i>GPT-2</i> (SV1: 27.2%)				
L3	96.1	10.2	0.222	−0.01
L12	77.8	35.6	0.256	+0.20
<i>Pythia-70M</i> (SV1: 92.1%)				
L3	67.8	6.6	0.073	+0.05
L6	6.1	3.8	0.968	+11.1
<i>Pythia-160M</i> (SV1: 90.6%)				
L12	7.7	5.7	0.995	+8.3

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